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# Correct Scores, Wrong Scientific Conclusions: Auditing Learned Surrogates of Chaotic Dynamics

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## Abstract

1        Learned forecasters increasingly stand in for scientific models that are themselves  
2        imperfect, and a forecast score alone cannot say whether a surrogate succeeds  
3        for the right reasons. We test five conditions that connect a score to a scientific  
4        claim (held-out generalization, complete failure accounting, matched disclosure,  
5        structural identifiability, and cross-system transfer) on  $k$ -pendulum ( $k \in \{1, 2, 3\}$ )  
6        and Lorenz–63 forecasts from language models, a time-series foundation model,  
7        learned dynamics, sparse identification, and numerical integrators. Every test alters  
8        or overturns a conclusion the raw comparison invites: the in-sample/out-of-sample  
9        gap is fourfold, two exact symmetries make absolute parameter recovery impossible,  
10       and a second system reverses the model ordering. On video-tracked hardware the  
11       roles invert: the RK4 model with published constants is itself imperfect, and its  
12       lead over simple baselines is gone within a second. A correct leaderboard can still  
13       support incorrect scientific conclusions.

## 14    1    Introduction

15       Scientific claims require more than low forecast error. When a learned forecaster stands in for an  
16       imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched  
17       inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of the  
18       surrogate. We study these failure modes directly, predicting a chaotic system’s state  $s_T$  from its state  
19        $s_0$  on two systems chosen to disagree: the  $k$ -link pendulum, chaotic for  $k \geq 2$ , with high-precision  
20       ground truth and control over what physical metadata a model is told, and Lorenz–63, which differs  
21       in observation model and hypothesis class; a hardware replication of the audit (§4.5) then scores the  
22       reference model itself as the imperfect one.

23       Our contribution is a claim audit for dynamical-system forecasts: held-out generalization, complete  
24       failure accounting, matched disclosure on identical trajectories, identifiability checked before scoring  
25       recovery, and transfer to a second system. Table 1 states the conclusion before and after each test;  
26       all five alter or reject a conclusion the unaudited comparison supports. These are conclusions the  
27       field draws: zero-shot forecasters are routinely scored without failure accounting or matched controls  
28       [Gruver et al., 2023, Zhang and Gilpin, 2025]; ours mirrors that practice. Three of the checks are the  
29       familiar concerns of imperfect models applied to the evaluation itself: misspecification (a Hamiltonian  
30       prior on damped dynamics), identifiability (exact parameter degeneracies), and transfer (a hypothesis  
31       class containing one system and not the other).

32       Appendix D situates the audited families and background.

Table 1: Conclusions before and after the claim audit.

| Test            | Unaudited conclusion                             | Audited conclusion                                                |
|-----------------|--------------------------------------------------|-------------------------------------------------------------------|
| Failures        | Answered forecasts represent model performance.  | Incomplete coverage lowers reliability-adjusted performance.      |
| Generalization  | Learned dynamics forecast accurately.            | At this budget, error rises about fourfold; simple baselines win. |
| Disclosure      | Withholding constants improves forecasts.        | Matched: small, model-dependent (+0.069 rad pooled).              |
| Identifiability | Absolute constants can be recovered from angles. | Two exact symmetries make absolute recovery impossible.           |
| Transfer        | The pendulum order ranks model families.         | Lorenz-63 produces a different order.                             |

## 33 2 Benchmark design

### 34 2.1 Task and systems

35 The pendulum state is  $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ ; hiding its constants does not change  
36 the trajectory, which permits the matched test in §4.2. Ground truth integrates the full nonlinear  
37 Lagrangian equations of motion with fixed-step RK4 at  $\Delta t = 10^{-3}$  s, recorded every 0.01 s plus a  
38 10 s pre- $t=0$  context for the time-series models. Initial conditions are sampled as  $\theta_i \sim \mathcal{U}(-3, 3)$  rad,  
39  $\omega_i \sim \mathcal{U}(-1, 1)$  rad/s with a fixed seed; the labels come from RK4, so its coarser-step variant is a  
40 reference by construction, not a competitor.

### 41 2.2 Regimes and grid

42 Three regimes test generalization: normal ( $g=9.81$ , undamped), changed\_disclosed ( $g=3.71$ ,  
43 damped), and changed\_hidden ( $g=1.62$ , damped, constants withheld; full values in Table 9). An  
44 evaluation cell is a tuple (model,  $k$ , regime, horizon), with horizons  $\{0.01, 1, 10, 60\}$  s. The LLM  
45 grid also varies modality and prompting; all comparisons here use the coordinate modality with direct  
46 prompting.

47 **Out-of-sample protocol.** Learned models are never scored on their training data: a shared held-out  
48 set (a separate seed) of 180 initial conditions (60 per regime,  $k \in \{1, 2, 3\}$ ) is scored identically for  
49 every family. The cross-family comparison uses horizons  $\{1, 10\}$  s, where chaos has begun but error  
50 has not saturated ( $180 \times 2 = 360$  cells per model); in-sample error appears once, to quantify the  
51 generalization gap (§4.1).

### 52 2.3 Inputs and grouping

53 Each family sees different information; the asymmetry is part of the experimental record (Table 3,  
54 Appendix B). The integrators get the exact equations and constants; the learned models get target-  
55 system training plus the constants (normal-regime priors in hidden cells); the LLMs get a text  
56 description with the constants except in hidden cells; Chronos-2 gets a 10 s pre- $t=0$  history window  
57 and no constants at all. The LLMs are large hosted models, while the learned models are small  
58 MLPs (Appendix B); larger physics-informed networks are out of scope. We separate known-physics  
59 methods (RK4, Euler, symplectic), which serve as numerical references, from the black-box methods  
60 being compared.

### 61 2.4 Metrics

62 The primary metric is mean absolute angle error (rad), wrapped to  $[-\pi, \pi]$  and averaged over links  
63 and trajectories. A cell is answered only if the model returns a parseable, finite prediction; because  
64 scoring answered cells alone would reward abstention on hard cells, unanswered cells receive the  
65 random-guess baseline  $\pi/2$  (the expected wrapped-angle error of a uniform guess), so answering is  
66 part of the score.

67 **Verification record.** Each cell retains its identifiers, prompt, raw response, parser outcome, pre-  
68 diction, token counts, and score. Local methods are reproducible from the released checkpoints and

seeds; hosted deployments can change or be withdrawn, so for them the released per-cell responses are the record that keeps the evaluation auditable.

**Disclosure gap.** We define  $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ ; positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, §4.2 also toggles disclosure on identical trajectories.

### 3 Methods

**LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt and JSON schema (Appendix A); each cell is one temperature-0 call with uniform budgets, no best-of- $N$ .

**Time-series foundation models.** We run Chronos-2 [Ansari et al., 2024] locally in its standard history-conditioned form: a 10 s pre- $t=0$  context of all  $2k$  channels forecasts to the target horizon, in univariate and multivariate modes (the paired mode comparison uses 20 trajectories per cell).

**Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`; we compare derivative-matching, rollout, and mixed-window rollout objectives (Appendix B). The primary models train on nine trajectories, one per  $(k, \text{regime})$  cell; a nested budget study trains fresh checkpoints at  $\{1, 5, 20\}$  trajectories per eligible cell with three initializations (HNN: undamped cells only, since damping violates Hamiltonian conservation).

**Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at  $\Delta t=0.01$  s using the exact equations of motion; their error is from discretization alone.

**Baselines.** Four baselines share the grid: persistence ( $\theta_T=\theta_0$ ), constant velocity ( $\theta_T=\theta_0+\omega_0 T$ ), a small-angle linearized model, and a nearest-neighbor analog forecast built from the learned models’ training data. On both held-out seeds (equal-weighted; §4.3) we also fit SINDy (degree-two, STLSQ) and NVAR (two delays, degree-two, ridge); divergent rollouts are failures at  $\pi/2$ .

**Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence, SINDy, NVAR, a Neural ODE trained on 20 single-seed trajectories, and zero-shot Chronos-2 on five five-trajectory held-out seeds, reporting training-normalized coordinate MAE at  $\{0.1, 0.5, 1.0\}$  s (configuration in Appendix C). HNN and LNN are omitted because Lorenz–63 is dissipative and non-Hamiltonian.

## 4 Results

### 4.1 Failure accounting and held-out generalization

Every model is scored on the shared held-out grid of 360 cells (180 trajectories, horizons  $\{1, 10\}$  s); the Answered column of Table 2 gives the fraction with valid predictions. `grok` answers 82% and `l1nn` 94%; their ranks fall when missing cells receive  $\pi/2$ , and a mean over answered cells alone would flatter both. We report 95% bootstrap intervals over trajectories (LLM per-call variation excluded).

Table 2 ranks the models under the access conditions of §2.3; the ranking is conditional on those inputs, this training budget, and this system (§4.4 tests transfer). The numerical integrators sit at the top; every learned-dynamics model lands closer to the  $\pi/2$  baseline than to the RK4 and symplectic references. The lowest-error black-box entry, the LLM `kimi` (0.538 rad, 99% answered), has an interval that only marginally excludes the nearest baseline’s: a statement about one recorded June 2026 deployment, not about model families; the Lorenz test rejects the family reading.

Under the equalized hidden-regime run, nearest-neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972), though those two intervals overlap. The stable learned models score about 0.26 rad on their training trajectories and 0.97–1.14 rad out of sample, so in-sample evaluation understates error about fourfold.

Table 2: Out-of-sample cross-family comparison (360 cells, horizons  $\{1, 10\}$  s): reliability-adjusted mean angle error (rad), unanswered cells at  $\pi/2$ , 95% bootstrap CI. RK4 generates the data (reference). †: configurations first evaluated in this work; ‡: simple baselines.

| #  | Model                     | Err   | 95% CI         | Answered | Family      |
|----|---------------------------|-------|----------------|----------|-------------|
| –  | rk4 (reference)           | 0.059 | [0.032, 0.089] | 100%     | numerical   |
| 1  | symplectic                | 0.205 | [0.163, 0.251] | 100%     | numerical   |
| 2  | euler                     | 0.405 | [0.344, 0.470] | 100%     | numerical   |
| 3  | kimi-k2.6                 | 0.538 | [0.475, 0.602] | 99%      | LLM         |
| 4  | chronos2-multi†           | 0.684 | [0.617, 0.752] | 100%     | time-series |
| 5  | chronos2-uni†             | 0.697 | [0.631, 0.766] | 100%     | time-series |
| 6  | nearest-neighbor‡         | 0.725 | [0.657, 0.797] | 100%     | baseline    |
| 7  | linearized‡               | 0.855 | [0.774, 0.939] | 100%     | baseline    |
| 8  | grok-4-1-fast-reasoning   | 0.881 | [0.805, 0.957] | 82%      | LLM         |
| 9  | deepseek-v4-pro           | 0.952 | [0.881, 1.027] | 100%     | LLM         |
| 10 | hnn-rollout               | 0.972 | [0.898, 1.047] | 100%     | learned     |
| 11 | persistence‡              | 1.004 | [0.934, 1.078] | 100%     | baseline    |
| 12 | neural-ode-rollout-mixed† | 1.036 | [0.967, 1.104] | 100%     | learned     |
| 13 | hnn-rollout-mixed†        | 1.066 | [0.990, 1.143] | 100%     | learned     |
| 14 | neural-ode                | 1.110 | [1.039, 1.181] | 100%     | learned     |
| 15 | neural-ode-rollout        | 1.121 | [1.054, 1.190] | 100%     | learned     |
| 16 | hnn                       | 1.138 | [1.059, 1.218] | 100%     | learned     |
| 17 | constant-velocity‡        | 1.319 | [1.241, 1.396] | 100%     | baseline    |
| 18 | lnn†                      | 1.356 | [1.282, 1.430] | 94%      | learned     |

113 SINDy (0.883 rad) and NVAR (1.248) appear only on the two-seed replication subset (Table 4);  
 114 SINDy lands between Chronos-2 and the learned models, NVAR near constant velocity.

## 115 4.2 Matched disclosure and identifiability

116 On the held-out set the hidden regime shows lower error than the disclosed ( $-0.390$  rad,  $p_{\text{Holm}} < 10^{-3}$ ,  
 117 unpaired) and normal ( $-0.524$ ) regimes; but the regimes use different trajectories, and the hidden  
 118 ones are slower and less divergent. Toggling disclosure on identical trajectories for the three LLMs  
 119 (900 paired cells on a separate matched grid; Table 8) reverses the pooled effect: hiding the constants  
 120 increases error ( $\Delta_{\text{disc}} = +0.069$  rad,  $p_{\text{Holm}} = 0.034$ ), with a strong penalty at  $k=1$  and a small reverse  
 121 effect at  $k=3$ ; the unadjusted advantage is a difficulty confound.

122 The disclosure test also has a structural limit: the angular equations are unchanged by  $(m, d) \mapsto$   
 123  $(am, ad)$  and  $(L, g, d) \mapsto (bL, bg, b^2d)$ , so for  $k=1$  only  $g/L$  and  $d/(mL^2)$  are identifiable, and  
 124 absolute  $g, L, m, d$  cannot be scored as independently recoverable. A single-state constant-naming  
 125 probe mostly returns Earth-standard defaults (a prior, not recovery; transcripts released); a numerical  
 126 check verifies both invariances to below  $10^{-13}$ .

## 127 4.3 Replication across seeds and training budgets

128 On an independent held-out seed (five trajectories per cell versus 20; Table 4), the selected ordering  
 129 is unchanged: numerical references lowest, Chronos-2 ahead of the learned models, HNN-rollout  
 130 best among them. Two seeds do not estimate seed-level uncertainty. The LNN’s low primary-seed  
 131 coverage was a hardware artifact: its per-step Hessian exhausted GPU memory on long rollouts. A  
 132 CPU rerun recovers the dropped cells ( $66\% \rightarrow 94\%$ ; both seeds use it); the remaining failures are  
 133 genuine divergences, scored at  $\pi/2$ , and the two-seed mean stays 1.366 rad.

134 **Training-budget check.** Across three initializations and both held-out seeds, Neural ODE error  
 135 falls from 1.090 at one trajectory per cell to 0.861 at twenty, and normal-regime HNN error from  
 136 1.269 to 0.981 (Table 10). The rows cover different regimes; more data helps the Neural ODE  
 137 monotonically, the HNN plateaus by 15 trajectories, and neither closes the gap to known physics.

## 138 4.4 Cross-system test: Lorenz–63

139 We test whether the pendulum conclusions transfer to Lorenz–63, a chaotic system with a polynomial  
 140 vector field. They do not. The identifiability verdict flips: on the pendulum the physical constants

are not recoverable from an angle trajectory (§4.2), while on Lorenz, SINDy recovers the governing equations themselves, not merely accurate forecasts. The refit selects exactly the true sparsity pattern and returns  $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$  with maximum coefficient error  $2.2 \times 10^{-13}$  (Appendix C); forecast error stays below  $2 \times 10^{-5}$  in every seed. The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is fully observed; the pendulum’s absolute constants are unidentifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, and the check makes it explicit before recovery is scored.

The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25 trajectories), a Lorenz Neural ODE reaches normalized MAE  $0.014 \pm 0.008$ ,  $0.079 \pm 0.030$ , and  $0.277 \pm 0.063$  at 0.1, 0.5, and 1.0 s (mean  $\pm$  SE over seeds), below both Chronos-2 modes at every horizon (multivariate  $0.101 \pm 0.018$ ,  $0.691 \pm 0.129$ ,  $0.771 \pm 0.097$ ; univariate similar); NVAR drifts to 0.429 by 1.0 s and persistence trails (0.900; Table 4). This reverses the pendulum result, where trained models lose to simple baselines: here the system-trained model beats the zero-shot foundation model. The families again see different inputs, so this is not an architecture race; what the cross-system result rejects is the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable seed: the Neural ODE beats both Chronos-2 modes in all 15 seed-by-horizon comparisons, and the identifiability result is not a sampling claim, since the symmetries are exact and Lorenz-63 lies in SINDy’s library by construction. The test excludes the API LLMs, HNN, and LNN (a Hamiltonian surrogate is misspecified for the dissipative flow); those omissions limit what it can establish, not what it can refute, and the Lorenz ordering is not offered as the correct one.

#### 4.5 Real data: the reference model becomes the imperfect one

On real hardware RK4 is no longer ground truth but the imperfect model. We use an open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published point-mass constants,  $\sim 0.35^\circ$  median tracking noise); Trial 1 serves only for calibration (nearest-neighbor database, SINDy fit, damping fit), and Trials 2–3 supply 120 held-out states scored as before at  $\{0.1, 0.5, 1.0\}$  s (Table 5, Appendix E).

RK4 with the published constants leads at 0.1 s (0.084 rad, about  $14\times$  the tracking noise) but reaches 1.25 rad by 1.0 s, indistinguishable from persistence (1.23); unmodeled friction and the point-mass simplification consume its advantage within a second. The fitted damping trades horizons (0.89 vs. 0.82 at 0.5 s, 1.15 vs. 1.25 at 1.0 s). SINDy, exact on Lorenz, sits at baseline level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells: states shortly after the drop lack its 5 s context (Appendix E), and failure accounting scores them at  $\pi/2$ . Every check transfers, and the ordering differs from both synthetic orderings.

#### 4.6 Diagnostics

Appendix F reports the remaining diagnostics. Two matter most: the physical-fidelity check localizes a misspecification (a symplectic HNN rollout leaves energy drift unchanged, so the fault is the learned Hamiltonian, not the integrator), and every black-box family saturates near the chaotic noise floor by 1 s, hence the  $\{1, 10\}$  s window. Error rises with  $k$  for all but one model; the hidden regime is easier for most models (the §4.2 confound); rollout loss is not uniformly better; and Chronos-2’s two modes are indistinguishable ( $n=720$ ).

### 5 Discussion and conclusion

Performance depends on system, observations, hypothesis class, data, and access, which a scalar leaderboard does not encode: each of the five checks changes a claim it supports, so benchmarks should report the checks, training budgets, and information access with the result. The synthetic systems make each check’s effect visible against exact ground truth; the hardware check runs the audit where the reference model itself is imperfect; all records needed to repeat it accompany the paper. **Limitations.** The LLM grid uses one held-out seed; the Lorenz test has five trajectories per seed, the hardware check three trials of one apparatus, SINDy needs the true degree in its library, and foundation-model corpora may contain pendulum-like data.

## References

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research (TMLR)*; arXiv:2403.07815, 2024.
- Johannes G. Borgqvist, Alexander P. Browning, Fredrik Ohlsson, and Ruth E. Baker. Framing global structural identifiability in terms of parameter symmetries. *arXiv preprint arXiv:2410.03757*, 2024.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15): 3932–3937, 2016. doi: 10.1073/pnas.1517384113.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. *ICLR 2020 Workshop on Integration of Deep Neural Models and Differential Equations*; arXiv:2003.04630, 2020.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2310.10688.
- Daniel J. Gauthier, Erik Bollt, Aaron Griffith, and Wendson A. S. Barbosa. Next generation reservoir computing. *Nature Communications*, 12:5564, 2021. doi: 10.1038/s41467-021-25801-2.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2110.05266.
- Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.07820.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.14858.
- Audun D. Myers, Joshua R. Tempelman, David Petrushenko, and Firas A. Khasawneh. Low-cost double pendulum for high-quality data collection with open-source video tracking and analysis. *HardwareX*, 8: e00138, 2020. doi: 10.1016/j.ohx.2020.e00138. Dataset: Mendeley Data, doi:10.17632/z4hvxjgtbz.2.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02592.
- Xin Xu, Qiyun Xu, Tong Xiao, Tianhao Chen, Yuchen Yan, Jiaxin Zhang, Shizhe Diao, Can Yang, and Yang Wang. Ugphysics: A comprehensive benchmark for undergraduate physics reasoning with large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.00334.
- Yuanzhao Zhang and William Gilpin. Zero-shot forecasting of chaotic systems. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2409.15771.

## A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

Table 3: Inputs available to each family. State is at  $t=0$ ; history is pre- $t=0$ . Chronos-2 is history-conditioned. \*In changed\_hidden cells the learned models receive normal-regime prior constants instead of the true ones.

| Family                       | State | History          | Constants               | Equations   |
|------------------------------|-------|------------------|-------------------------|-------------|
| Numerical (RK4/Euler/sympl.) | ✓     | —                | ✓ (all regimes)         | ✓ (exact)   |
| Learned (NODE/HNN/LNN)       | ✓     | —                | ✓ base; withheld*       | — (learned) |
| Time-series (Chronos-2)      | —     | ✓ (10 s context) | —                       | —           |
| LLMs (kimi/grok/deepseek)    | ✓     | —                | ✓ except changed_hidden | —           |

### 237 System prompt (direct / no-CoT).

238 You are a careful physicist predicting the future state of a  $k$ -pendulum (a chain of  
 239  $k$  point masses on massless rods, hinged in a planar chain, swinging under gravity  
 240 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,  
 241 no code fences. Schema:  
 242 {"theta": [...k floats...], "omega": [...k floats...]}  
 243 theta\_i is in radians (link i measured from straight-down,  
 244 positive counter-clockwise)  
 245 and omega\_i is angular velocity in rad/s.

246 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,  
 247 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a  
 248 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}.”

### 249 User prompt (coordinate modality).

250 Number of links  $k = \{k\}$   
 251 Physical constants:  
 252  $g$  (gravity,  $\text{m/s}^2$ ):  $\{g\}$   
 253  $L$  (rod lengths,  $\text{m}$ ):  $\{L\}$   
 254  $m$  (point masses,  $\text{kg}$ ):  $\{m\}$   
 255 damping (viscous coefficient on omega):  $\{\text{damping}\}$   
 256 Initial state at  $t=0$ :  
 257 theta\_0 (rad):  $\{\text{theta}_0\}$   
 258 omega\_0 (rad/s):  $\{\text{omega}_0\}$   
 259 Predict the state at  $t = \{T\}$  seconds.  
 260 Return theta and omega as length- $\{k\}$  JSON arrays.

261 **Hidden-constants regime.** The constants block above is replaced by:

262 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point  
 263 masses, and damping are NOT disclosed. You must infer dynamics from the initial  
 264 conditions alone.

265 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the  
 266 `<answer>...</answer>` tags; the first  $\{\dots\}$  object is JSON-parsed and validated to contain length- $k$  theta  
 267 and omega arrays. A response that fails any check is counted as unanswered and scored at  $\pi/2$ .

## 268 B Training and compute

269 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate,  
 270 and train on a single 8 GB consumer GPU. Mixed- $k$  rollout adds negligible parameters; its cost is wall-clock  
 271 time (the HNN mixed-window run is the most expensive at  $\sim 6.7$  h because each unroll step recomputes  $\nabla H$   
 272 via autograd). Chronos-2 runs at inference only. The dataset has 67,491 samples across  $k \in \{1, 2, 3\}$  and  
 273 three regimes, subsampled at  $\Delta t=0.01$  s from the  $10^{-3}$  s ground truth. Despite the sample count, the training  
 274 set contains only *nine* trajectories, one for each  $(k, \text{regime})$  cell. The budget study uses nested subsets of 1,  
 275 5, and 20 trajectories per eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or  
 276 180 trajectories; HNN sees 3, 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds  
 277 throughout.

**Training objectives.** Derivative matching minimizes MSE on finite-difference  $\dot{s}$ ; rollout loss minimizes MSE on the state after  $w$  Euler steps with gradients through the unroll; mixed-window rollout trains on several window lengths jointly (Neural ODE:  $w \in \{10, 50\}$ ; HNN:  $w \in \{5, 20\}$ ). The LNN’s loss requires second-order automatic differentiation; the LNN trains at full scale (67k samples, hidden 256) on a CUDA GPU. Chronos-2 resamples the context to at most 512 points and forecasts at most 64 steps.

**Reproducibility records.** The anonymized artifact accompanying this submission contains the code, prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep; the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at temperature 0. The reported hosted-model numbers are recomputed from the saved records; re-calling the API is a new evaluation rather than a reproduction. Invalid cells are scored at  $\pi/2$ .

## C Seed and system replication

The Lorenz–63 test uses  $(\sigma, \rho, \beta) = (10, 28, 8/3)$ , RK4 ground truth at  $\Delta t=0.01$  s, a 5 s context, one fixed training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity pattern exactly and recovers  $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$  with maximum absolute coefficient error  $2.2 \times 10^{-13}$ . NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and their seed mean; horizons  $\{1, 10\}$  s. The primary seed has 20 trajectories per  $(k, \text{regime})$  cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz–63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below  $2 \times 10^{-5}$  in every cell. Lower is better.

| Pendulum model           | Primary seed  | Second seed   | Seed mean           |
|--------------------------|---------------|---------------|---------------------|
| rk4 (reference)          | 0.059         | 0.074         | 0.067               |
| symplectic               | 0.205         | 0.194         | 0.200               |
| euler                    | 0.405         | 0.415         | 0.410               |
| chronos2-uni             | 0.697         | 0.604         | 0.650               |
| chronos2-multi           | 0.684         | 0.621         | 0.653               |
| sindy                    | 0.860         | 0.906         | 0.883               |
| hnn-rollout              | 0.972         | 0.858         | 0.915               |
| neural-ode-rollout-mixed | 1.036         | 0.989         | 1.012               |
| hnn-rollout-mixed        | 1.066         | 1.037         | 1.051               |
| nvar                     | 1.245         | 1.251         | 1.248               |
| lnn                      | 1.356         | 1.376         | 1.366               |
| Lorenz model             | 0.1 s         | 0.5 s         | 1.0 s               |
| rk4 (reference)          | 0.000         | 0.000         | 0.000               |
| sindy                    | $<10^{-5}$    | $<10^{-5}$    | $<2 \times 10^{-5}$ |
| neural-ode               | 0.014 (0.008) | 0.079 (0.030) | 0.277 (0.063)       |
| nvar                     | 0.012 (0.001) | 0.262 (0.050) | 0.429 (0.084)       |
| chronos2-uni             | 0.126 (0.027) | 0.700 (0.102) | 0.732 (0.085)       |
| chronos2-multi           | 0.101 (0.018) | 0.691 (0.129) | 0.771 (0.097)       |
| euler                    | 0.028 (0.005) | 0.216 (0.023) | 0.885 (0.120)       |
| persistence              | 0.524 (0.054) | 1.028 (0.123) | 0.900 (0.057)       |

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## D Related work

The audited models span learned dynamics [Chen et al., 2018, Greydanus et al., 2019, Cranmer et al., 2020] on chaotic testbeds [Gilpin, 2021], zero-shot LLM and foundation-model forecasting [Gruver et al., 2023, Zhang and Gilpin, 2025, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], LLM physical reasoning [Lewkowycz

Table 5: Real-data check: mean wrapped angle error (rad) over both links on Trials 2–3 (120 held-out states), failures at  $\pi/2$ . The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor  $\sim 0.006$  rad.

| Model                          | 0.1 s | 0.5 s | 1.0 s | Answered |
|--------------------------------|-------|-------|-------|----------|
| physics-rk4                    | 0.084 | 0.820 | 1.248 | 100%     |
| physics-rk4-damped             | 0.086 | 0.892 | 1.145 | 100%     |
| nearest-neighbor (cross-trial) | 0.371 | 0.849 | 1.117 | 100%     |
| chronos2                       | 0.611 | 1.160 | 1.223 | 89%      |
| constant-velocity              | 0.345 | 1.402 | 1.576 | 100%     |
| sindy-real                     | 0.490 | 1.501 | 1.535 | 99%      |
| persistence                    | 0.934 | 1.654 | 1.230 | 100%     |

et al., 2022, Wei et al., 2022, Xu et al., 2025], and sparse identification [Brunton et al., 2016, Gauthier et al., 2021]. The identifiability check draws on structural identifiability under parameter symmetries [Borgqvist et al., 2024]. Unlike in selective prediction, which reports risk together with coverage [Geifman and El-Yaniv, 2017], failed cells here stay in the main score because models are compared at fixed task coverage.

## E Real-data check details

The hardware dataset [Myers et al., 2020] is released under GPLv3 with video tracking of both link angles at 500 Hz and per-frame uncertainty estimates (median  $0.20^\circ$  upper link,  $0.43^\circ$  lower). We use the published simplified point-mass parameters ( $L = [0.172, 0.143]$  m,  $m = [0.311, 0.111]$  kg,  $g = 9.81$  m/s<sup>2</sup>). Angular velocities come from a Savitzky–Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the pendulum positioned by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation states are drawn evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame, leaving a gravity-referenced offset per link that we select from  $\{0, \pi\}$  by whichever choice lets the reference model track at a 20 ms horizon; the selected offsets are zero for every trial, with 20 ms error of 0.005–0.007 rad against a 0.006 rad tracking floor. A grid search over  $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$  selects damping  $10^{-3}$  on Trial 1 at the 0.5 s horizon. SINDy uses a degree-two polynomial plus first-order Fourier library on Trial 1 with Savitzky–Golay derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz and forecasts at most 50 steps. Failures are scored at  $\pi/2$ ; Chronos-2’s are input-infeasible cells (insufficient context) rather than divergences, and the fixed-coverage rule scores both alike.

## F Extended diagnostics

This appendix expands the diagnostics summarized in §4.6.

**Behavior by horizon.** Out-of-sample error by horizon appears in Table 13. At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed-window Neural ODE is intermediate at both horizons (0.951/1.120). Figure 1 shows the growth curves over the  $\{0.01, 1, 10, 60\}$  s ladder: the numerical integrators stay near zero until  $\sim 10$  s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The  $\{1, 10\}$  s window used for the comparison is the range over which the families remain separable.

**Breakdown by  $k$  and regime.** Tables 6 and 7 report out-of-sample error by number of links and by regime. Error increases with  $k$  for every model except the mixed-window Neural ODE, which dips at  $k=3$  (for kimi,  $0.11 \rightarrow 0.55 \rightarrow 0.95$ ; the integrators are near-exact at  $k=1$  and degrade as chaos sets in), and the cross-family ordering holds within each  $k$ . The `changed_hidden` regime has the lowest error for most models (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the constants reduces but does not remove the difficulty effect: hidden-regime errors are now 0.71–1.19; the earlier values favor models given the true constants. The low-gravity hidden constants ( $g=1.62$ ) produce slower, less-divergent motion over a fixed horizon, so the unadjusted hidden-regime advantage is a difficulty confound (§4.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than  $\sim 0.03$  rad and leaves the ranking unchanged.

**Training-objective check.** Table 11 reports the effect of the loss with architecture fixed. Under the equalized hidden-regime run, rollout loss is not uniformly better: mixed-window rollout is lower for the Neural ODE, while single- $k$  rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model

Table 6: Out-of-sample error by number of links  $k$ .

| Model                    | $k=1$ | $k=2$ | $k=3$ |
|--------------------------|-------|-------|-------|
| rk4 (reference)          | 0.000 | 0.068 | 0.109 |
| symplectic               | 0.000 | 0.253 | 0.363 |
| euler                    | 0.271 | 0.410 | 0.534 |
| kimi-k2.6                | 0.112 | 0.553 | 0.949 |
| chronos2-multi           | 0.263 | 0.798 | 0.992 |
| chronos2-uni             | 0.279 | 0.813 | 0.999 |
| grok-4-1-fast-reasoning  | 0.366 | 1.030 | 1.247 |
| deepseek-v4-pro          | 0.872 | 0.949 | 1.036 |
| hnn-rollout              | 0.524 | 0.985 | 1.406 |
| neural-ode-rollout-mixed | 0.845 | 1.178 | 1.084 |
| hnn-rollout-mixed        | 0.550 | 1.162 | 1.487 |
| neural-ode               | 0.626 | 1.304 | 1.400 |
| neural-ode-rollout       | 0.968 | 1.172 | 1.224 |
| hnn                      | 0.511 | 1.337 | 1.567 |
| lnn                      | 1.072 | 1.440 | 1.556 |

error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN is numerically unstable and generalizes poorly out of sample.

**Physical-fidelity check.** Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 12 reports mean absolute drift over 10 s, normalized by  $E^* = g \sum_i m_i L_i$ , and the fraction of bounded rollouts. The learned models do not match numerical integrators on this axis: the lowest-drift learned model is 0.779, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so changing the integrator does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs. 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

**Multivariate time-series check.** On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- $k$  trend in the primary grid.

**Predictability horizon.** Pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold  $\tau$  from  $t=0$ ; the contiguous rule is used because, after divergence, a later horizon can fall below  $\tau$  by angle-wrap coincidence. Table 14 sweeps  $\tau$ .

## G Breakdowns by link count and regime

Reliability-adjusted mean angle error (rad) on the shared held-out grid (horizons  $\{1, 10\}$  s), by number of links (Table 6) and by regime (Table 7); models in the order of Table 2. Cells a model never produced (e.g. the LNN’s diverged  $k=3$ ) are scored at  $\pi/2$ . Each row uses the same grid as Table 2.

## H Additional result tables

Full per-model values for the horizon breakdown and the predictability horizon (both defined in Appendix F), together with the regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

Table 7: Out-of-sample error by regime. Learned-model hidden cells use the equalized hidden-regime run. The hidden regime remains easier for most models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §4.2 consequently mixes disclosure with difficulty.

| Model                    | normal | disclosed | hidden |
|--------------------------|--------|-----------|--------|
| rk4 (reference)          | 0.174  | 0.003     | 0.000  |
| symplectic               | 0.387  | 0.177     | 0.051  |
| euler                    | 0.743  | 0.387     | 0.085  |
| kimi-k2.6                | 0.635  | 0.519     | 0.461  |
| chronos2-multi           | 0.818  | 0.826     | 0.409  |
| chronos2-uni             | 0.818  | 0.798     | 0.475  |
| grok-4-1-fast-reasoning  | 1.096  | 0.959     | 0.588  |
| deepseek-v4-pro          | 1.240  | 1.097     | 0.520  |
| hnn-rollout              | 1.087  | 1.115     | 0.713  |
| neural-ode-rollout-mixed | 1.260  | 0.836     | 1.011  |
| hnn-rollout-mixed        | 1.244  | 1.172     | 0.782  |
| neural-ode               | 1.221  | 1.095     | 1.014  |
| neural-ode-rollout       | 1.281  | 0.894     | 1.189  |
| hnn                      | 1.262  | 1.303     | 0.850  |
| lnn                      | 1.525  | 1.416     | 1.127  |

Table 8: Matched disclosure gap (rad), paired on identical trajectories.  $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ ; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models  $\times$  2 horizons  $\times$  150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to those in Table 7. Per-model  $\Delta$  values are computed before rounding.

| Model                     | Disclosed | Hidden | $\Delta_{\text{disc}}$ |
|---------------------------|-----------|--------|------------------------|
| kimi-k2.6                 | 0.746     | 0.957  | +0.211                 |
| grok-4-1-fast-reasoning   | 0.987     | 1.025  | +0.038                 |
| deepseek-v4-pro           | 1.425     | 1.382  | −0.042                 |
| pooled (paired, $n=900$ ) |           |        | +0.069*                |
| $k=1$                     |           |        | +0.297***              |
| $k=3$                     |           |        | −0.055*                |

\* $p_{\text{Holm}} < 0.05$ , \*\*\* $p_{\text{Holm}} < 10^{-3}$ .

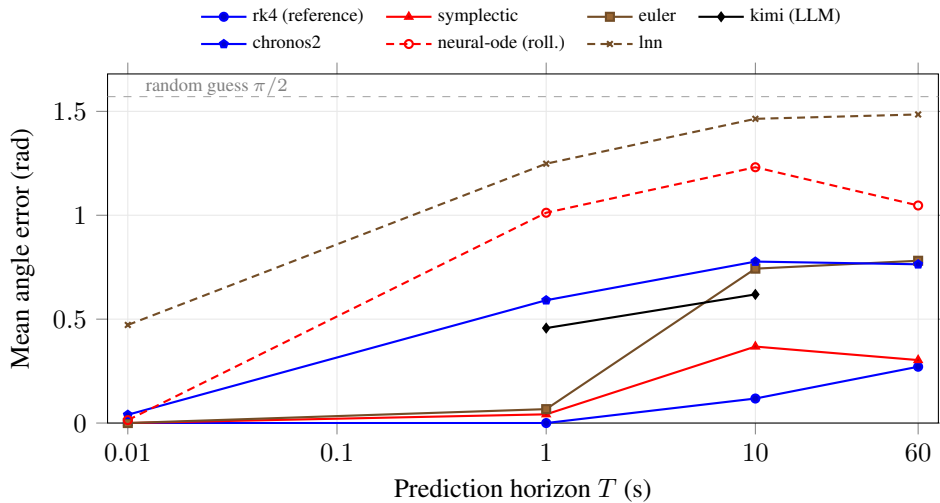


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until  $\sim 10$  s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs are evaluated at  $\{1, 10\}$  s only. Dashed line: random-guess baseline  $\pi/2$ .

Table 9: Physical regimes. The changed\_hidden regime withholds the constants.

| Regime            | $g$ (m/s <sup>2</sup> ) | $L$ (m)     | $m$ (kg)    | damping | disclosed |
|-------------------|-------------------------|-------------|-------------|---------|-----------|
| normal            | 9.81                    | 1.0         | 1.0         | 0.0     | —         |
| changed_disclosed | 3.71                    | 1.4/0.7/1.1 | 2.0/0.6/1.5 | 0.05    | yes       |
| changed_hidden    | 1.62                    | 0.8/1.2/0.9 | 0.5/1.8/1.0 | 0.10    | no        |

Table 10: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals come from hierarchical resampling over initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

| Model      | Scope       | Total trajectories | Error | 95% CI         |
|------------|-------------|--------------------|-------|----------------|
| HNN        | normal      | 3                  | 1.269 | [1.209, 1.349] |
| HNN        | normal      | 15                 | 0.974 | [0.891, 1.062] |
| HNN        | normal      | 60                 | 0.981 | [0.856, 1.113] |
| Neural ODE | all regimes | 9                  | 1.090 | [1.011, 1.207] |
| Neural ODE | all regimes | 45                 | 0.968 | [0.923, 1.028] |
| Neural ODE | all regimes | 180                | 0.861 | [0.829, 0.907] |

Table 11: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons  $\{1, 10\}$  s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ( $n=720$ , Wilcoxon+Holm).

| Model      | Objective              | Err   | vs. prev. row                                  |
|------------|------------------------|-------|------------------------------------------------|
| Neural ODE | derivative MSE         | 1.110 | —                                              |
|            | rollout (single- $k$ ) | 1.121 | n.s. ( $p_{\text{Holm}}=0.188$ )               |
|            | mixed-window           | 1.036 | sig. ( $p_{\text{Holm}}=0.013$ )               |
| HNN        | derivative MSE         | 1.138 | —                                              |
|            | rollout (single- $k$ ) | 0.972 | sig. ( $p_{\text{Holm}}=1.95 \times 10^{-4}$ ) |
|            | mixed-window           | 1.066 | sig. higher ( $p_{\text{Holm}}=0.014$ )        |

Table 12: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by  $E^*=g \sum_i m_i L_i$  (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

| Model                    | Family    | Energy drift | Stable |
|--------------------------|-----------|--------------|--------|
| rk4                      | numerical | 0.000        | 100%   |
| symplectic               | numerical | 0.066        | 100%   |
| euler                    | numerical | 0.494        | 100%   |
| neural-ode-rollout-mixed | learned   | 0.779        | 100%   |
| hnn-rollout-mixed        | learned   | 0.865        | 100%   |
| neural-ode-rollout       | learned   | 0.872        | 100%   |
| hnn-rollout              | learned   | 2.065        | 100%   |
| lnn                      | learned   | 4.038        | 27%    |
| hnn                      | learned   | 4.270        | 100%   |
| neural-ode               | learned   | 4.346        | 100%   |

Table 13: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons  $\{1, 10\}$  s.

| Model                    | 1 s   | 10 s  |
|--------------------------|-------|-------|
| rk4 (reference)          | 0.000 | 0.118 |
| symplectic               | 0.042 | 0.368 |
| euler                    | 0.067 | 0.743 |
| kimik2.6                 | 0.457 | 0.619 |
| chronos2-multi           | 0.591 | 0.777 |
| chronos2-uni             | 0.622 | 0.773 |
| grok-4-1-fast-reasoning  | 0.905 | 0.857 |
| deepseek-v4-pro          | 0.996 | 0.909 |
| hnn-rollout              | 0.855 | 1.088 |
| neural-ode-rollout-mixed | 0.951 | 1.120 |
| hnn-rollout-mixed        | 0.982 | 1.150 |
| neural-ode               | 0.892 | 1.328 |
| neural-ode-rollout       | 1.012 | 1.231 |
| hnn                      | 1.052 | 1.224 |
| lnn                      | 1.248 | 1.464 |

Table 14: Predictability horizon (s): longest horizon tracked contiguously below threshold  $\tau$ , on the shared out-of-sample set (ladder  $\{0.01, 1, 10, 60\}$  s). Higher is better; “—” means not even the first horizon stays below  $\tau$ .

| Model                    | $\tau=0.1$ | 0.25 | 0.5  | 1.0  |
|--------------------------|------------|------|------|------|
| rk4 (reference)          | 1.0        | 10   | 60   | 60   |
| symplectic               | 1.0        | 1.0  | 60   | 60   |
| euler                    | 1.0        | 1.0  | 1.0  | 60   |
| neural-ode-rollout       | 0.01       | 0.01 | 0.01 | 0.01 |
| neural-ode-rollout-mixed | 0.01       | 0.01 | 0.01 | 1.0  |
| chronos2-multi           | 0.01       | 0.01 | 0.01 | 60   |
| chronos2-uni             | 0.01       | 0.01 | 0.01 | 60   |
| hnn-rollout              | 0.01       | 0.01 | 0.01 | 1.0  |
| neural-ode               | 0.01       | 0.01 | 0.01 | 1.0  |
| hnn                      | 0.01       | 0.01 | 0.01 | 0.01 |
| lnn                      | —          | —    | 0.01 | 0.01 |